This is the authors' version of the following article: Lee, H., & Lee, J. H. (2022). The effects of robot-assisted language learning: A meta-analysis. *Educational Research Review, 35,* 100425. The version of record is available at https://doi.org/10.1016/j.edurev.2021.100425. This version differs from the published article in layout and pagination; please cite the published version.

The effects of robot-assisted language learning: A meta-analysis

Hansol Lee[a], Jang Ho Lee[b,*]

[a] *Department of English, Korea Military Academy, Seoul, South Korea*
[b] *Department of English Education, Chung-Ang University, Seoul, South Korea*

[*] Corresponding author. Chung-Ang University, 84 Heukseok-ro, Dongjak-gu, Seoul, 06974 , South Korea. *E-mail address:* jangholee@cau.ac.kr (J.H. Lee).

Abstract

Given the importance of learner–interlocutor interactions, social robots have received considerable attention in the field of language learning, as this technology can physically accompany learners, is equipped with common social behaviors, and can effectively communicate with learners for language acquisition. In line with the continuous efforts to examine the effects of using social robots in language learning, this study is a meta-analysis aiming to integrate the findings of previous experimental studies on robot-assisted language learning (RALL), focusing on language learning achievement. To this end, we selected 12 independent samples (N = 522) from the initial pool of 16 empirical studies—including 23 independent samples (N = 1147)—and computed effect sizes along with their standard errors. The results indicated that RALL had a medium-sized average effect (d = 0.59, SE = 0.09, 95% CI: 0.41–0.76) compared to non-RALL conditions, such as traditional learning conditions without technology (e.g., teacher-only) or other technology-mediated conditions (e.g., virtual agents and tablet PCs). We further identified several moderator variables (i.e., age group, target language, language domain, robots' role, interaction type, type of non-RALL condition), and provided an in-depth discussion on how these variables were related to the effectiveness of using robots for language learning.

Keywords: Human-computer interaction; Language learning; Meta-analysis; Robot-assisted learning; Social robots

1. Introduction

The value of interactions has been recognized in the scholarly branches of children's language acquisition (Hockema & Smith, 2009; Iverson, 2010; Wellsby & Pexman, 2014), second language (L2) acquisition (Long, 1996; Mackey, 2007), and sociocultural theory (Bruner,

1983; Vygotsky, 1980). Despite differences in the focuses of their research, these branches have collectively proposed that language acquisition begins in interactions with others. In addition, the knowledge emerging from such interactions is internally and cognitively processed in individuals' minds.

Furthermore, the importance of interaction has been extensively acknowledged in the frameworks of computer-assisted language learning (CALL; Chapelle, 1998, 2001). This has led researchers and material developers to consider this element in the development of CALL-based instructions. Consequently, CALL research has started to draw on technologies in computer science. This has enabled language learners to linguistically interact with technology, powered by machine and deep learning techniques, high-performance computing, natural language processing systems, and large-scale datasets, including chatbots (Fryer et al., 2020), machine reading comprehension (Liu et al., 2019; Zeng et al., 2020), and social robots (Lee & Lee, 2021; Randall, 2019; van den Berghe et al., 2019). Among these, social robots—the target technology of the present study—have recently received considerable attention because of their economic feasibility (Lee & Lee, 2021), the previously mentioned technological developments, and their potential to serve as effective mediums for interaction with language learners.

By imitating normal human-to-human interactions, social robots are "designed to interact and communicate with people" (van den Berghe et al., 2019, p. 260) through common social behaviors, such as languages, gestures, and facial expressions (Bartneck & Forlizzi, 2004). The phrase "social robots" is intrinsically descriptive. First, the word "social" narrows down the type of robot. In other words, social robots are different from general robots creating cars in factories because the former communicate with people. Second, the word "robot" refers to tangible machines existing in the same physical space as people, unlike animated characters that are designed for limited interaction with people through computer screens. Social robots have been widely used for educational purposes, such as teaching arithmetic and doing puzzles (Kennedy et al., 2015; Leyzberg et al., 2012).

Social robots have only recently been actively employed in language learning (e.g., Tsai, 2019; Vogt et al., 2019). However, their physical embodiment increases their value as pedagogical resources to be considered for language teaching and learning. This is because, unlike 2D technologies (e.g., animated characters or virtual agents on computers and tablets), social robots can manipulate objects, move their bodies, and express emotions through gestures and facial expressions. This indicates that social robots have a greater potential to be implemented under a wider range of pedagogical approaches to language teaching and learning compared to other technologies. By virtue of these features, social robots seem capable of effectively displaying immediacy (Randall, 2019), referring to communication behaviors oriented toward

bridging the psychological and physical distances between interlocutors (Mehrabian, 1968). In the field of education, teachers' immediacy behaviors can enhance learners' participation in class (Menzel & Carrell, 1999) and their affective and cognitive learning (Chesebro & McCroskey, 2001; Christensen & Menzel, 1998). In terms of social robots, such features can contribute to effective learning (Kennedy et al., 2015). Social robots, particularly anthropomorphic ones, tend to be considered as realistic conversation interlocutors, compared to the 2D technologies mentioned above, because of their physical embodiment, thereby enhancing learners' engagement in language learning (Chang et al., 2010). Younger learners may perceive social robots in the form of personified animals as familiar conversational partners (Bartneck et al., 2009; Beran et al., 2011), which increases their motivation to learn the target language.

In addition to these strengths, social robots have further advantages over human teachers. For example, they could provide an unlimited amount of language learning practice to learners, without any restrictions on time and place, or fatigue (Bushweller, 2020). They could also record and save each learner's utterances into their memory and provide more individualized linguistic feedback (Randall, 2019), which may be a challenging task for human teachers (see Lee & Lee, 2021 for a similar discussion). Finally, social robots could provide learning opportunities for children with chronic illness who cannot gain access to teachers' lessons (Newhart et al., 2017).

Two systematic reviews (Randall, 2019; van den Berghe et al., 2019) have recently been conducted to offer a qualitative summary of previous empirical studies on robot-assisted language learning (RALL). Randall (2019) reviewed a total of 79 RALL studies published between 2004 and 2017. Her review addressed the following features: the characteristics of the social robots used in the studies (e.g., function, form, voice, role, verbal and non-verbal interaction, signal, and personification), education effects by language domain (e.g., vocabulary, grammar, pronunciation, speaking, reading comprehension, and writing), changes in affective variables on the part of learners (e.g., motivation, interest, participation, confidence, and anxiety), and learning contexts (e.g., learners' age, language proficiency, target language, instructional type, location, and duration). According to the review, social robots have a potential to improve learners' language acquisition and create positive changes in learners' emotional domains (e.g., lowering the language learning anxiety).

With a similar goal, van den Berghe et al. (2019) reviewed 33 studies that focused on using social robots to improve vocabulary, reading, speaking, grammar, and sign language skills. The findings were mixed regarding social robots' effects on language learning. The results of studies on improving conversation skills and vocabulary yielded both positive and negative findings, whereas the learners were found to benefit from RALL-based conditions in terms of

improving reading, grammar, and sign language skills. In response to these complex results, van den Berghe et al. (2019) suggested that to accurately examine the effects exerted by social robots on language learning, future research should investigate the characteristics of RALL, such as the roles in which social robots are the most effective, the age group best suited to social robots' effective functioning, and the frequency most suitable for RALL.

In line with van den Berghe et al. (2019) and Randall (2019), the present study aims to examine the overall effectiveness of RALL through a meta-analysis. This study investigates whether the RALL condition is more beneficial than the non-RALL condition by aggregating the experimental results of published empirical studies, after converting them into comparable effect sizes. Only a few empirical studies with relatively small sample sizes have examined this topic. However, a meta-analysis can provide an integrated and comprehensive perspective with greater statistical power than a single, small empirical study. The present study aims to contribute to RALL by responding to van den Berghe et al.'s (2019) calls for future research, addressing the impact of empirically proven, effective learning conditions for RALL according to age, target language, language domain, role of robots, interaction type, and instruction duration, Thus, this meta-analysis was conducted for the following reasons: (1) to gauge the overall average effect of using robots on language learning, (2) to identify important moderator variables of RALL-based experimental studies, and (3) to estimate the extent to which these variables influenced the effects of using robots on language learning.

2. Methods

2.1. Literature search

We used the keywords "robot-assisted language learning (RALL)," "robot & language," "robot & learning," "robot & language learning," "robot & education," "robot & literacy," and "robot & teaching" to identify previous empirical studies published up to the year 2019, using databases such as Google Scholar, ProQuest, and Web of Science. Ten related journals (*Applied Learning, CALICO Journal, Computer-Assisted Language Learning, Computers & Education, Educational Research Review, Language Learning, Language Learning & Technology, ReCALL, Review of Educational Research,* and *System*) were perused. The empirical studies included in the previously mentioned systematic reviews on RALL were also reviewed and added to the literature search.

As shown in Fig. 1, the procedure identified 231 studies after the duplicates were deleted. The studies' titles and abstracts were reviewed, and 177 were found to be irrelevant to the topic (e.g., empirical studies without a language learning measure and review articles). The remaining 54 studies were thoroughly reviewed to ensure that they met the following five

inclusion criteria (1) the study should include a comparison (i.e., non-RALL): condition supporting the examination of the effects of using robots on language learning; (2) the study should report descriptive statistics necessary for effect-size calculations, pertaining to the effects of using robots on the achievements in language learning; (3) the study should assess outcomes related to language learning; (4) the study should ensure homogeneity between the groups assigned to the RALL and non-RALL conditions through either randomized controlled trials or the pretests, which could confirm the equivalence of students' prior language proficiency; and (5) the study should be written in English. A total of 38 studies were excluded because they did not meet these criteria; finally, 16 studies remained. The reasons for exclusion were as follows: (1) 16 studies lacked non-RALL conditions, (2) 6 did not report the necessary descriptive data, (3) 15 did not assess outcomes related to language learning, and (4) 1 was not written in English (although its title and abstract were in English).

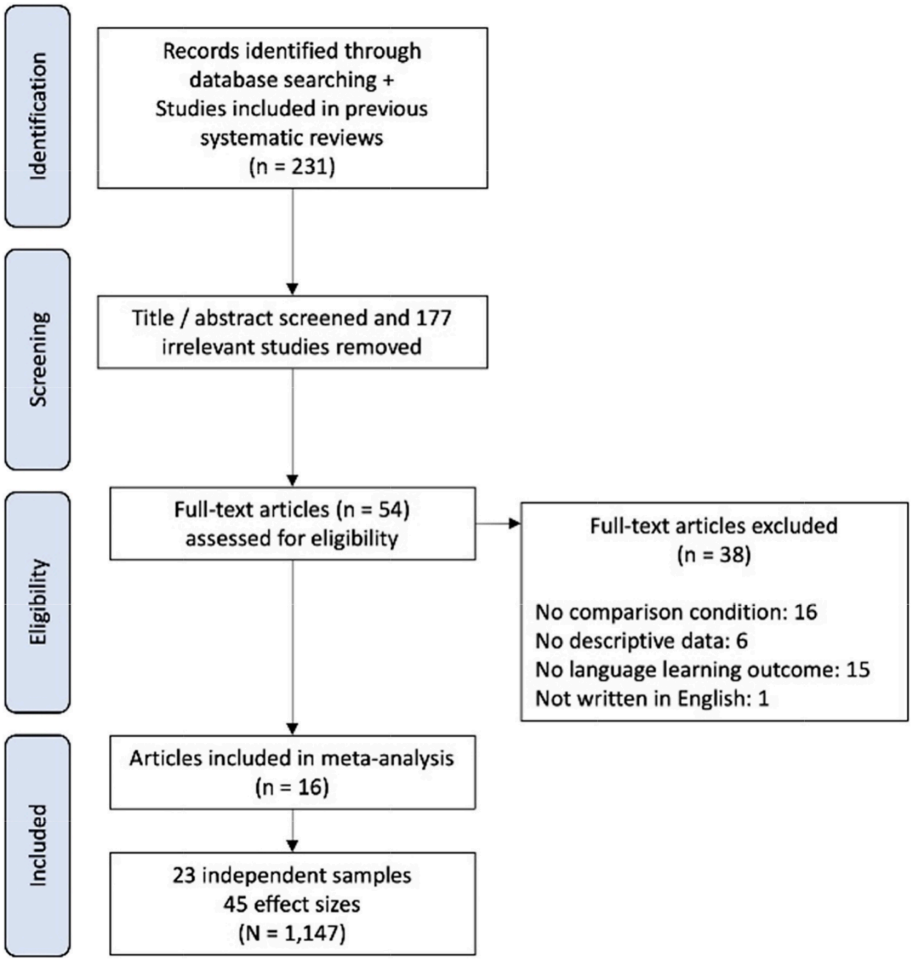


Fig. 1. PRISMA flowchart (Moher, Liberati, Tetzlaff, Altman, & PRISMA Group, 2009).

2.2. Effect-size calculation

While the identified experimental studies addressed the effects of using robots on language learning, these were different from each other in several aspects. In particular, the effects of RALL, as presented by each study, are difficult to compare objectively because of measurement differences regarding target language domains, testing types, and instructional instruments. Therefore, the quantifiable results of each intervention were converted into scale-free effect sizes; we used Cohen's unbiased *d* (also known as Hedges' *g*) computation in the present study.

Cohen's *d* is an effect-size measurement based on standardized mean differences, which is calculated by dividing the mean difference between the treatment and control groups by the pooled standard deviation. In addition, each effect size entails its standard error, which is calculated by multiplying the pooled standard deviation by the square root of the sum of the inverse of each sample (i.e., 1 divided by treatment group number + 1 divided by control group number). Cohen's unbiased *d* is a complement compensating for the vulnerability of Cohen's *d*, with respect to small sample sizes (typically, those that are smaller than 50; Hedges & Olkin, 1985).

Prior to calculating the effect size, we found that three studies in our dataset (Han et al., 2008; Köse et al., 2015; Wedenborn, 2015) had two independent samples. In addition, van den Berghe et al.'s (2018) study had two control conditions, with two independent comparisons. Further, the study by Vogt et al. (2019) had two treatment conditions, in which robots behaved differently, and two control conditions; therefore, a total of four independent samples (or comparisons) were identified. Notably, in the present meta-analysis study, we use the terms *sample* and *comparisons* interchangeably. Thus, we calculated 23 effect sizes for each independent sample distinguished from the 16 identified studies, with a total sample size of 1147 (Table 1).

Table 1
Sample size of each independent sample in the initial dataset.

Independent sample	Sample size	
	RALL group	Non-RALL group
Akalin et al., 2014	18	10
Alemi et al. (2014)	30	16
Eimler et al., 2010	9	9
Han et al. (2008) [1]	30	30
Han et al. (2008) [2]	30	30
Hong et al. (2016)	25	27
Hsiao et al. (2015)	30	27
Hyun et al. (2008)	17	17

Independent sample	Sample size	
	RALL group	**Non-RALL group**
Köse et al. (2015) [1]	12	12
Köse et al. (2015) [2]	8	8
Mazzoni and Benvenuti (2015)	4	6
Perlmutter et al. (2016)	12	12
Tsai (2019)	28	25
van den Berghe et al. (2018) [1]	23	23
van den Berghe et al. (2018) [2]	23	21
Vogt et al. (2019) [1]	54	54
Vogt et al. (2019) [2]	54	54
Vogt et al. (2019) [3]	54	32
Vogt et al. (2019) [4]	54	32
Wang et al. (2013)	32	31
Wedenborn (2015) [1]	15	15
Wedenborn (2015) [2]	15	15
Wu et al. (2015)	31	33
Total: 23 independent samples (k = 23)	608	539
	N = 1147	

After calculating effect sizes, we checked if there was any sign of publication bias—also known as small-study effects—which occurs when small studies with positive findings are more likely to be published for the topic (Lee, Chung, Zhang, Abedi, & Warschauer, 2020; Lee, Warschauer, & Lee, 2019). To this end, visual (e.g., funnel plot) and mathematical (e.g., Egger's test) checks were employed. Fig. 2 illustrates two funnel plots using different precision metric (i.e., standard error and the inverse standard error) for the initial dataset to implement the visual check. The two plots showed that there were about 11 dots outside the funnels.

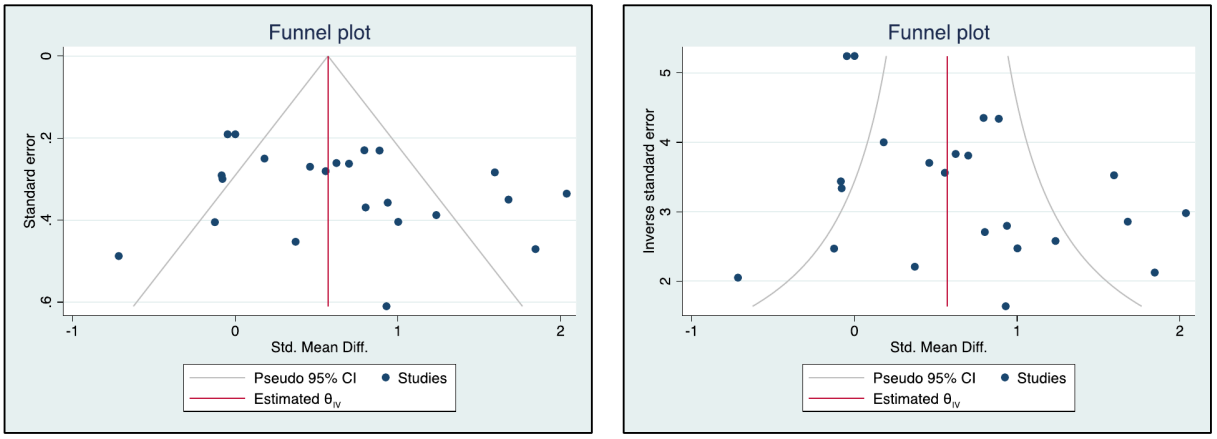


Fig. 2. Funnel plots for the initial dataset.

However, the results of the Egger's test showed a non-significant *p*-value (p = .359). Thus, we tried to measure the extent of heterogeneity within the initial dataset. The results indicated that the dataset had an I^2 of 79.73, and the *p*-value of the Chi-square test for homogeneity was less than .001. Given that I^2 is acceptable when it is less than 25%, and that the *p*-value of the Chi-square test is acceptable when it is larger than .05 (Higgins & Thompson, 2002), we concluded that the initial dataset included biased effect sizes and decided to drop the outliers from the dataset.

Fig. 3 showed that the revised dataset did not include any outliers according to the recalculated two funnel plots. Moreover, the results of the Egger's test showed a non-significant *p*-value (p = .792); in addition, the results of the heterogeneity tests indicated that the revised dataset had an I^2 of zero, and the *p*-value of the Chi-square test was 0.541. As shown in Table 2, in the revised dataset, we included 12 independent samples, with a total sample size of 522.

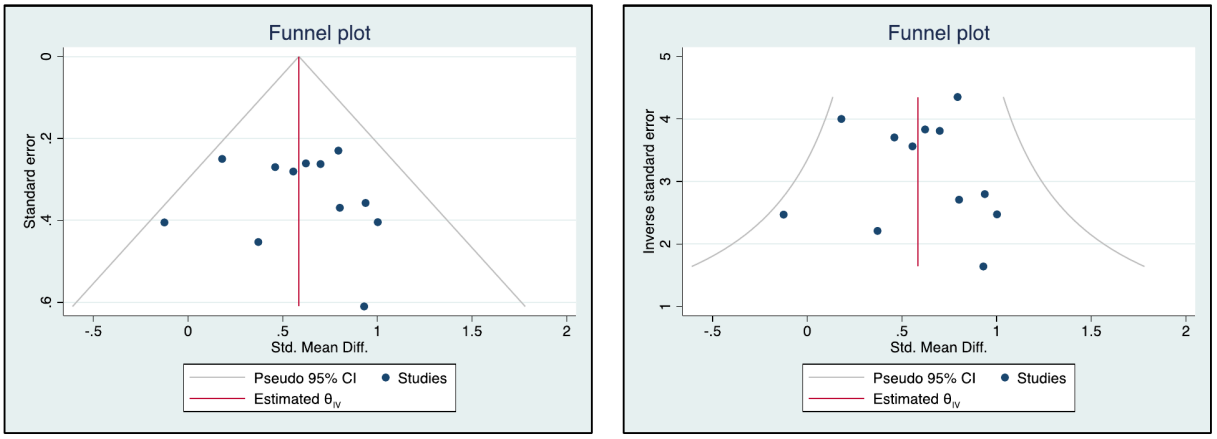


Fig. 3. Funnel plots for the revised dataset.

Table 2
Sample size of each independent sample in the revised dataset.

Independent sample	Sample size	
	RALL group	Non-RALL group
Akalin et al., 2014	18	10
Eimler et al., 2010	9	9
Han et al. (2008) [1]	30	30
Han et al. (2008) [2]	30	30
Hong et al. (2016)	25	27
Hsiao et al. (2015)	30	27
Hyun et al. (2008)	17	17
Mazzoni and Benvenuti (2015)	4	6
Perlmutter et al. (2016)	12	12
Vogt et al. (2019) [3]	54	32
Wang et al. (2013)	32	31

Independent sample	Sample size	
	RALL group	Non-RALL group
Wedenborn (2015) [1]	15	15
Total: 12 independent samples (k = 12)	276	246
	N = 522	

2.3. Coding scheme

Table 3
The specifications of the included samples in terms of the selected moderators.

Independent sample	Moderators						
	Age	Target language	Instruction duration	Language domain	Role(s) of robots	Interaction type	Non-RALL condition type
Akalin et al., 2014	10–12	Turkish sign language in Turkey	1 session	Vocabulary	Tutor	One-on-one	Virtual program
Eimler et al., 2010	9–11	English in Germany	20 min	Vocabulary	Tutor	One-on-one	Pencil-paper
Han et al. (2008) [1]	10–12	English in Korea	40 min	Overall	Tutor	One-on-one	Audio + book
Han et al. (2008) [2]	10–12	English in Korea	40 min	Overall	Tutor	One-on-one	Computer
Hong et al. (2016)	10–11	English in Taiwan	1 semester	Listening	TA[a]	One-on-one	Teacher
Hsiao et al. (2015)	2–3	Chinese in China	2 months	Overall	Tutor	In groups	Tablet PC
Hyun et al. (2008)	4	Korean in Korea	4 weeks	Reading	Tutor	One-on-one	Media-assisted
Mazzoni and Benvenuti (2015)	4–6	English in Italy	45 min	Vocabulary	Peer	One-on-one	Peers
Perlmutter et al. (2016)	Adult	Dutch, Swedish in US	45 min	Overall	Tutor	One-on-one	Web application
Vogt et al. (2019) [3]	5–6	English, Dutch in the Netherlands	3 weeks	Vocabulary	Tutor	One-on-one	Danced with robot
Wang et al. (2013)	10–11	English in Taiwan	1 semester	Speaking	Peer	In groups	Classmates
Wedenborn (2015) [1]	Adult	Russian in Sweden	15 min	Vocabulary	Tutor	One-on-one	Animated face on screen

[a] Teaching assistant.

Table 4
Coding results of the moderators.

Independent sample	Moderators						
	Age group	Target language	Instruction duration	Language domain	Role(s) of robots	Interaction type	Non-RALL condition type
Akalin et al., 2014	Upper elementary	L1 (Turkish sign language)	1 week	Vocabulary	T/T[a]	One-on-one	Other technology
Eimler et al., 2010	Upper elementary	L2 (English)	1 week	Vocabulary	T/T	One-on-one	Traditional learning condition with no technology
Han et al. (2008) [1]	Upper elementary	L2 (English)	1 week	Other	T/T	One-on-one	Other technology
Han et al. (2008) [2]	Upper elementary	L2 (English)	1 week	Other	T/T	One-on-one	Other technology
Hong et al. (2016)	Upper elementary	L2 (English)	15 weeks	Other	T/T	One-on-one	Traditional learning condition with no technology

Independent sample	Moderators						
	Age group	Target language	Instruction duration	Language domain	Role(s) of robots	Interaction type	Non-RALL condition type
Hsiao et al. (2015)	Kindergartener	L1 (Chinese)	10 weeks	Other	T/T	In groups	Other technology
Hyun et al. (2008)	Kindergartener	L1 (Korean)	4 weeks	Other	T/T	One-on-one	Other technology
Mazzoni and Benvenuti (2015)	Kindergartener	L2 (English)	1 week	Vocabulary	Peer	One-on-one	Traditional learning condition with no technology
Perlmutter et al. (2016)	Adult	L2s (Dutch, Swedish)	1 week	Other	T/T	One-on-one	Other technology
Vogt et al. (2019) [3]	Kindergartener	L2s (English, Dutch)	3 weeks	Vocabulary	T/T	One-on-one	Other technology
Wang et al. (2013)	Upper elementary	L2 (English)	15 weeks	Other	Peer	In groups	Traditional learning condition with no technology
Wedenborn (2015) [1]	Adult	L2 (Russian)	1 week	Vocabulary	T/T	One-on-one	Other technology

[a] Teaching assistant/Tutor.

A coding procedure was implemented to search for and identify comparable features of RALL among the 12 included samples. First, a list of anticipated features was designed based on the previous review studies. In the study by Randall (2019), experimental studies were analyzed according to features such as age group, the number of participants, the presence of a control group and posttests, instruction duration, and target language domains. In the study by van den Berghe et al. (2019), target language domains, the number of participants, experimental methods (with or without control groups), and types and roles of social robots were key features used to analyze the included previous studies. Further, the collected studies reported that they had different types of non-RALL conditions (e. g., teacher, peer, tablet, and virtual agent) and interactions with robots (i.e., one-on-one and in groups). Thus, a total of seven key features were determined as potential moderator variables for the current meta-analysis, namely (1) age groups, (2) target language, (3) instruction duration, (4) language domains, (5) social robots' roles, (6) interaction type, and (7) non-RALL condition type. For learners' language proficiency levels, we found that all studies included beginner level learners (regardless of the target language), except for one study (i.e., Akalin et al., 2014). Thus, we decided against including this feature as a moderator variable. Table 3 presents the seven moderator variables from the revised dataset. For our moderator analyses to investigate the ways in which these variables influence the effects of using robots on learners' language learning, we categorized the values for each variable into theoretically and statistically comparable codes. Table 4 presents the results of our coding for these variables.

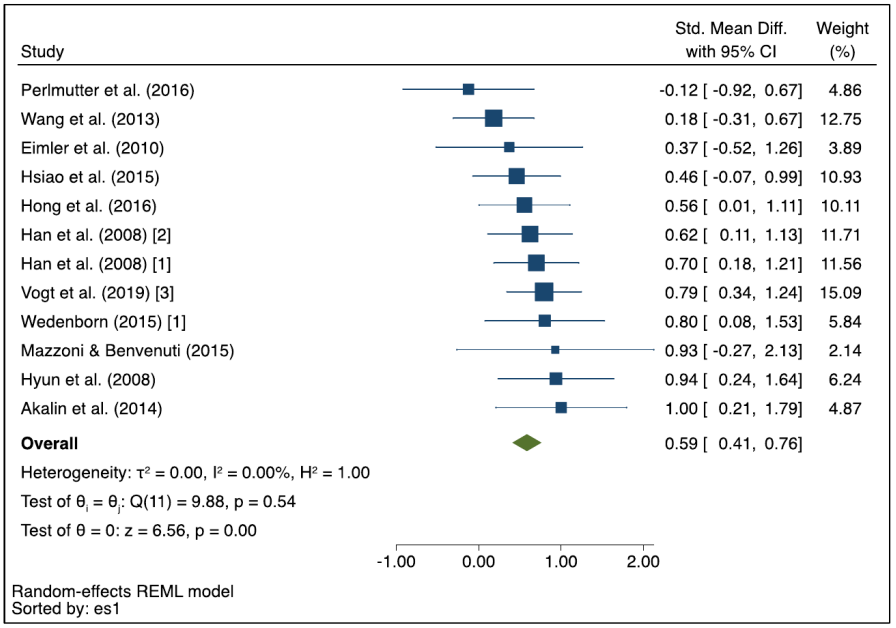


Fig. 4. Forest plot for the calculated effect sizes.

Table 5

Descriptive and inferential statistics for the moderator analysis.

Moderator variable	**Descriptive statistics**			**Meta-regression**		**Diff. at 5%**
	N	**Mean**	**SE**	**Mean**	**SE**	
Age group						n/a
Kindergartner	4	0.72***	(0.15)	0.59**	(0.19)	
Upper elementary	6	0.54***	(0.12)	0.65***	(0.14)	
Adult	2	0.38	(0.27)	0.27	(0.29)	
Target language						
L1	3	0.72***	(0.19)	0.80**	(0.25)	
L2	9	0.55***	(0.10)	0.53***	(0.11)	
Language domain						
Vocabulary	5	0.78***	(0.16)	0.77***	(0.20)	
Others	7	0.49***	(0.11)	0.50***	(0.12)	
Role(s) of robots						
TA/tutor	10	0.64***	(0.10)	0.54***	(0.11)	
Peer	20	0.29	(0.23)	0.86*	(0.42)	
Interaction type						
One-on-one	10	0.67***	(0.10)	0.74***	(0.14)	
In groups	2	0.31	(0.18)	0.10	(0.36)	
Non-RALL condition type						
Traditional condition without technology	4	0.39*	(0.17)	0.28	(0.35)	
Other technology	8	0.66***	(0.11)	0.71***	(0.17)	

Note. The duration variable was included as a continuous variable in the equation and its coefficient was not statistically significant ($p > .05$).

*$p < .05$, **$p < .01$, ***$p < .001$.

The following is a description of how the selected moderators have been coded. First, for the age group variable, we identified three different groups: preschoolers, upper elementary graders, and adults. As Randall (2019) found that young students (first-graders) interacted with and were more interested in robots than the older students (sixth-graders), we decided to divide them into three categories: kindergartner (ages 2–6), upper elementary students (ages 9–12), and adult learners. Second, for the target language variable, we only had two categories, namely first language (including sign language) and second language. Third, the instruction duration was coded as a continuous variable (i.e., weeks), providing a practical guideline to our readers. Fourth, regarding the language domain, because a large proportion of studies examined vocabulary as the target (5 out of 12), while other aspects (i.e., listening, reading, and speaking) lacked enough sample sizes to be recognizable categorical values (i.e., only one study for each domain), we categorized this moderator variable into "vocabulary" and "other aspects" (see van den Berghe et al., 2019 for a relevant approach). Fifth, regarding the roles of social robots, we followed each sample's description of their social robots' use: teaching assistant (TA; a robot accompanied teacher), tutor, peer, companion, and interlocutor. Furthermore, teaching assistants and tutors were similar because they provided specific types of instruction to students. When it comes to peers, companions, and interlocutors, it was found that they generally helped learners' language learning through cooperative and horizontal relationships as peers. Therefore, we decided to categorize these values into two: TA/tutor vs. peer. Sixth, there were two interaction types with social robots in the sampled studies: "one-on-one" and "in groups." Lastly, we acknowledged the variation across the non-RALL conditions, and decided to categorize the differences into two: (1) traditional learning conditions without technology (e.g., teacher-only) and (2) learning conditions with other technologies (e.g., tablet PC).

2.4. Data analysis

To analyze the effectiveness of RALL, it is necessary to calculate the overall estimates by aggregating the calculated effect sizes. It is also necessary to calculate how the selected moderator variables influence the effectiveness of RALL. Finally, it should be noted that the calculated effect sizes are statistical estimates from different sample sizes, implying that the critical value of each effect size depends on its standard errors. Therefore, effect sizes with smaller standard errors should be given more weight than those with higher standard errors, when calculating the overall average effect size and utilizing statistical models, such as regression analysis (see Lee et al., 2019 for the rationale). To this end, this study used STATA 14 (StataCorp, 2015) and its command *meta* to calculate the overall average of effect sizes and

conduct a meta-regression analysis (i.e., including moderator variables in the equation) to investigate how the moderator variables influenced the effects of using robots on language learning.

3. Results

3.1. General analyses

The overall average effect size was 0.59 (*SE* = 0.09; $p < .001$; 95% CI: 0.41–0.76), after considering the standard error (weights) of each effect size and the seven moderator variables (see 3.2. *Moderator analyses* for the results related to the moderators). This overall average effect size can be considered medium-sized, according to Cohen's effect-size interpretation benchmarks (0.2, 0.5, and 0.8, as small, medium, and large, respectively). Thus, using social robots had a positive effect on language learning; on average, its effect was greater than that of non-RALL conditions. Fig. 4 shows the forest plot for the calculated effect size of each study.

3.2. Moderator analyses

Regarding moderator variables' influence on the effectiveness of RALL, Table 5 presents the results of descriptive statistics as well as a meta-regression for the six identified moderator variables, except for the instruction duration variable included as a continuous variable in the equation, without any significant contribution to the overall effectiveness. The left side of Table 5 shows the descriptive statistics, including the average effect sizes by the values of each moderator variable and the number of associated effect sizes. The right side of the table presents the results of the meta-regression, in which the seven moderator variables were included as independent variables in the model; the reported values for each moderator variable should be interpreted after holding the other variables constant. The results presented in Table 5 indicate that, on average, RALL exhibited positive treatment effects across different moderator variables (i.e., *d*s > 0); however, there was no statistically significant difference between the categorical values for each moderator variable (i.e., *p*s > .05).

Regarding the age group and target language related to learner factors, the left and right sides of Table 5 show mixed patterns across variables. With respect to age groups, there were medium-sized effects on the language learning of kindergarteners and upper elementary graders ($d = 0.72$, $p < .001$; $d = 0.54$, $p < .001$) and small-sized effects for adult learners ($d = 0.38$, $p > .05$). After controlling for the other moderator variables, the meta-regression results revealed that the effects of RALL were larger for upper elementary graders and kindergartners ($d = 0.65$, $p < .001$; $d = 0.59$, $p < .01$) than adult learners ($d = 0.27$, $p > .05$); however, there was no statistically significant difference ($p > .05$). Regarding the target language, there were

medium-sized effects on L1 (d = 0.72, p < .001) and L2 learning (d = 0.55, p < .001). The meta-regression produced a similar pattern. However, there was no statistically significant difference between these two values.

Regarding the role of social robots, it was found that average effect sizes were 0.64 (p < .001) and 0.29 (p > .05) for the experiments using social robots as teaching assistants/tutors and peers, respectively. However, the results of the meta-regression revealed a different pattern; that is, when controlling for the other moderator variables, the results of the meta-regression showed an average effect-size estimate of 0.54 (p < .001) and 0.86 (p < .05) for the experiments in which social robots were used as teaching assistants/tutors and peers, respectively, albeit no statistically significant difference.

For the remaining three variables, the left and right sides of Table 5 present similar patterns. In terms of the language domain, when dividing the calculated effect sizes into those conducted to improve vocabulary versus other aspects, the average effect-size estimates were 0.78 (p < .001) and 0.49 (p < .001), respectively. When all the effect sizes were included in the meta-regression, with the other moderator variables as independent variables, the results showed similar effect size estimates, albeit no statistically significant difference. For the interaction type, providing one-on-one learning opportunities with social robots (d = 0.67, p < .001) showed a larger impact than having students in groups with the social robots (d = 0.31, p > .05), and the effect-size gap between these two values increased after controlling for other factors (d = 0.74, p < .001 for one-on-one; d = 0.10, p > .05 for in group), again albeit no statistically significant difference. Notably, the effect size of using social robots became marginal (i.e., d < 0.20) only when students were in groups for RALL. In the case of the non-RALL condition, using social robots to supplant the other technology (d = 0.66, p < .001) seems to be more useful than providing the robots as an additional technology to traditional learning conditions without technology (d = 0.39, p < .05), and this pattern remained valid (d = 0.71, p < .001 for supplanting other technology; d = 0.28, p > .05 for providing an additional technology) after controlling for other features, even though there was no statistically significant difference between the effect-size estimates.

4. Discussion

The present study aimed to investigate the overall average effect of RALL and understand how different moderators influence the effectiveness. The main findings of the current meta-analysis are as follows:

(1) RALL has shown positive treatment effects on language learning when compared to non-RALL conditions.

(2) Regarding the identified moderator variables related to learner factors and intervention features, RALL has shown positive treatment effects across studies, regardless of these variables.

(3) Regarding moderators, RALL showed a larger effect size for kindergartners and upper elementary learners than for adult students, and for learning first languages (including sign languages) than learning second languages. In addition, RALL had a larger average effect size for teaching vocabulary than other language aspects. RALL also showed different effect sizes among the roles that robots adopted, interaction types, and non-RALL condition types. However, there was no statistically significant difference between the categorical values for the aforementioned moderator variables.

Based on these findings, an overall interpretation of the effectiveness of RALL and its pedagogical implications are discussed in Section 4.1.

4.1. The effectiveness of robot-assisted language learning

The positive average effect size (d = 0.59), which was computed based on the 12 independent samples (N = 522), indicated a positive pedagogical effect compared to non-RALL conditions. This result arose from between-group comparisons (i.e., treatment vs. control conditions) rather than from within-group comparisons (i.e., changes in treatment conditions only), as well as from statistically controlling for the identified moderator variables, considering the methodological differences across the collected studies. Thus, the resultant effect-size estimates from the current meta-analysis may yield more reliable and accurate inferences, addressing the call from the previous systematic review (van den Berghe et al., 2019).

This finding coincides with a convention that such technologies enabling linguistic interaction with human users could supplement language learning by offering effective learning opportunities (e.g., Atwell, 1999; Chapelle, 2001). The positive findings related to the RALL may be attributed to several characteristics of social robots: (1) their physical embodiment (Lee & Lee, 2021; van den Berghe et al., 2019) differentiating them from virtual characters on the screen; (2) ability to show verbal and non-verbal immediacy, enhancing language learning (Randall, 2019); (3) interaction competence through which they provide language input, negotiate meaning with learners, and elicit learners' output; and (4) social behaviors facilitating interaction between the robots and learners (Bartneck & Forlizzi, 2004).

Notably, however, that the result related to this overall effectiveness of using robots for language learning is qualified by the methodological limitations of the previous studies, which generally did not employ randomized controlled trials (although they checked the homogeneity between the RALL and non-RALL conditions through the pre-test; e.g., Hsiao et

al., 2015). Even those that employed random assignments of the participants did not statistically confirm whether these assignments were successful in ruling out any differences among groups (although participants were randomly chosen and subsequently assigned to the groups; e.g., Alemi et al., 2014). Notably, the related experiments tended to have small sample sizes because of practical considerations of bringing a limited number of social robots to classrooms. For example, among the 23 independent samples, more than half (12 samples) involved fewer than 50 learners; meanwhile, the largest independent sample consisted of 108 learners at the most (Vogt et al., 2019). This attribute, which highlights the lack of a large-scale field study on this topic, may be one of the reasons why previous reviews (Randall, 2019; van den Berghe et al., 2019) failed to draw a comprehensive conclusion. Given these methodological limitations, future studies should employ randomized controlled trials, include a follow-up confirmation procedure, and design their experiments such that there is no difference between groups other than the use of social robots. Future studies should also include large sample sizes to guarantee compelling statistical power.

4.2. The evidenced-based suggestions for robot-assisted language learning based on the moderator analyses

In the present study, we identified seven moderator variables and investigated how they influenced the effectiveness of RALL. This was in response to van den Berghe et al.'s (2019) calls for future research on how social robots could be used to maximize language learning. Based on the present study's findings, the following suggestions are proposed for effectively utilizing social robots in future language learning contexts.

First, regarding the age groups, kindergartner and upper elementary learners seemed to benefit more than adult learners from RALL. This result may be due to the previous research findings that young learners generally find RALL to be engaging and motivating (e.g., Han et al., 2008; Hong et al., 2016; Tanaka & Matsuzoe, 2012) and the proposition that "social robots could provide unique support" for such learner population (Kanero et al., 2018, p. 147) based on the features of social robots mentioned in Section 1 (i.e., *Introduction*). Thus, language educators for young learners are encouraged to consider this finding and employ social robots in their lessons if their pedagogical context permits. Nevertheless, our dataset included only three age groups, and the question of whether RALL would be effective for secondary-level learners—who are chronologically between these age groups—remains unanswered.

Second, social robots seemed to be more effective than non-RALL conditions for both L1 and L2 learning, but with a slightly more positive result for the former. Arguably, the acquisition of L2 is more complex than that of L1 (Ortega, 2009), resulting from additional factors such as the existence of another language (i.e., L1) in the learners' language system, the learner's age

(i.e., L2 learners tend to be older), and their motivation to learn the language. Thus, this result implies that more variables are involved in RALL in L2 learning, which future studies are encouraged to address. Nevertheless, our results regarding this moderator suggest that social robots could be employed in both L1 and L2 learning, indicating that these have a good potential as an educational resource.

Third, instruction duration was not a significant variable, according to the result of our meta-regression model. Thus, our study did not observe that social robots in language learning may have a short-term novelty effect (e.g., Fernaeus et al., 2010), when the durations were coded in terms of the weeks. Thus, although more studies, particularly longer ones, varying in terms of the duration related to the implementation of RALL are needed (see Randall, 2019 for a similar point), our result tentatively suggests that learners may benefit from robot-based instruction, regardless of the instruction duration. This finding may be used by language educators in their implementation of social robots.

Fourth, regarding the language domains, the findings of our moderator analysis showed a slightly more positive result for vocabulary than the “other” category, after controlling for the other moderating variables via meta-regression. Vocabulary, compared to other language aspects, is easier to be individually and independently presented (Lee & Lee, 2021); thus, it appears that the pedagogical usefulness of RALL in teaching vocabulary is indicated. However, it should be noted that, due to the small sample sizes, some language domains—for which social robots may be potentially effective in terms of language learning outcomes—were subsumed into the “other” category. Thus, we cannot rule out the possibility that social robots may be more effective for some language domains than for vocabulary, which should be addressed in future research.

Fifth, when controlling for the other moderator variables, it was found that social robots showed a larger effect size when taking on the role of peers rather than teaching assistants/tutors. In view of this finding as well as the existing discomfort associated with robots playing the role of teachers in class (e.g., Osada et al., 2006), we suggest that robots be employed as peers who are preset to be on par with students, or those who are slightly more advanced, to provide some type of linguistic scaffolding (Bruner, 1983) for a wide range of interactive classroom tasks. Alternatively, they may occasionally serve as assistants to human teachers by taking on a limited role in lessons (see Alemi et al., 2014), thereby making these robots less intimidating (rather than replacing human teachers).

Sixth, the result related to interaction type revealed that social robots may introduce better learning outcomes when employed in a “one-on-one” type, rather than in a group. This is expected, as social robots in such an interaction pattern may provide language learners with a

larger amount of linguistic interaction that is considered an essential element of language learning (Long, 1996; Mackey, 2007). In addition to the result associated with the role of social robots, language educators are then encouraged to assign the role of peers or alike to this technological resource and employ it in pair-based language tasks or activities.

Finally, between the two types of comparison (i.e., non-RALL) conditions, it was the other technology with which RALL showed a larger effect size than traditional learning conditions without technology. This finding indicates that social robots may have some potential to replace other technologies such as tablet PCs, web application, and virtual agents, but less so with the traditional learning environments involving human teachers or peers. Although more studies are needed, this finding indicates the potential of RALL over other existing technologies and the importance of improving social robot technology to match human interactions in language learning.

5. Conclusion and future directions

The present study ascertained the effectiveness of RALL through a meta-analysis and investigated how its effectiveness was influenced by the selected moderators. We found that RALL exerted a positive pedagogical effect on language learning, compared to the non-RALL condition, in addition to the positive treatment effects, regardless of these moderators. With a total sample size of 522 learners, the results of the current meta-analysis have generated sufficient statistical power; however, given that there were only 12 independent samples in the revised dataset and each sample had about 50 learners on average, future research should obtain more samples to conduct further experimental studies (see van den Berghe et al., 2019 for a similar suggestion).

In addition to large-scale studies, three promising avenues for experimental studies on the effects of using robots on language learning include the following. First, as robots may record the status of each learner's interlanguage and the progress on a target activity (Lee & Lee, 2021), the effect of robots' personalized or individual feedback on language learning may be a pedagogically important topic to pursue. Second, the effects of diverse types of verbal and non-verbal immediacy on robots taking on the roles of teaching assistants or tutors should be examined (Randall, 2019), as these variables may enhance language learners' cognitive and affective learning. Finally, future efforts on this issue should be geared toward exploring the effects of robots for enhancing language learners' oral (speaking) competence, in addition to considering the interaction effects between this language domain and different roles adopted by robots, given the potential of robots as real conversational partners in class (Chang et al., 2010). Future experimental studies addressing these suggestions may help expand our

understanding of the effectiveness of RALL and be used to verify the findings of the current meta-analysis. This could, in turn, be included in follow-up meta-analyses aimed toward accumulating significant and cogent RALL-related knowledge.

Author statement

Hansol Lee: Conceptualization, Methodology, Data curation, Writing-Original draft preparation and Editing.

Jang Ho Lee: Conceptualization, Supervision, Writing-Original draft preparation, Reviewing, and Editing.

Acknowledgements

This article was supported by 2020 research fund (20-군학-7) of Korea Military Academy (Hwarangdae Research Institute).

References

References marked with an asterisk indicate studies included in meta-analysis.

* Akalin, N., Uluer, P., & Kose, H. (2014). Non-verbal communication with a social robot peer: Towards robot assisted interactive sign language tutoring. In 2014 *IEEE-RAS international conference on humanoid robots* (pp. 1122–1127). IEEE.

* Alemi, M., Meghdari, A., & Ghazisaedy, M. (2014). Employing humanoid robots for teaching English language in Iranian junior high-schools. *International Journal of Humanoid Robotics, 11*(3), 1450022. https://doi.org/10.1142/S0219843614500224.

Atwell, E. (1999). *The language machine: The impact of speech and language technologies on English language teaching.* British Council.

Bartneck, C., & Forlizzi, J. (2004). A design-centred framework for social human-robot interaction. In *Proceedings of the Ro-Man 2004* (pp. 591–594). IEEE. https://doi.org/10.1109/ROMAN.2004.1374827.

Bartneck, C., Kulić, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International Journal of Social Robotics, 1*, 71–81. https://doi.org/10.1007/s12369-008-0001-3

Beran, T. N., Ramirez-Serrano, A., Kuzyk, R., Fior, M., & Nugent, S. (2011). Understanding how children understand robots: Perceived animism in child–robot interaction. *International Journal of Human-Computer Studies, 69*, 539–550. https://doi.org/10.1016/j.ijhcs.2011.04.003

* van den Berghe, R., van der Ven, S., Verhagen, J., Oudgenoeg-Paz, O., Papadopoulos, F., & Leseman, P. (2018). Investigating the effects of a robot peer on L2 word learning. In *Companion of the 2018 ACM/IEEE international conference on human-robot interaction* (pp. 267–268). ACM. https://doi.org/10.1145/3173386.3176986.

van den Berghe, R., Verhagen, J., Oudgenoeg-Paz, O., van der Ven, S., & Leseman, P. (2019). Social robots for language learning: A review. *Review of Educational Research, 89*(2), 259–295. https://doi.org/10.3102/0034654318821286

Bruner, J. (1983). *Child's talk: Learning to use language.* Oxford University Press.

Bushweller, K. (2020, January 7). Teachers, the robots are coming. But that's not a bad thing. *Education Week.* https://www.edweek.org/technology/teachers-the-robots-are-coming-but-thats-not-a-bad-thing/2020/01.

Chang, C.-W., Lee, J.-H., Chao, P.-Y., Wang, C.-Y., & Chen, G.-D. (2010). Exploring the possibility of using humanoid robots as instructional tools for teaching English as a second language in primary school. *Educational Technology & Society, 13*(2), 13–24. https://www.jstor.org/stable/jeductechsoci.13.2.13.

Chapelle, C. (1998). Multimedia CALL: Lessons to be learned from research on instructed SLA. *Language, Learning and Technology, 2*(1), 22–34. https://lib.dr.iastate.edu/engl_pubs/50/.

Chapelle, C. A. (2001). *Computer applications in second language acquisition.* Cambridge University Press.

Chesebro, J. L., & McCroskey, J. C. (2001). The relationship of teacher clarity and immediacy with student state receiver apprehension, affect, and cognitive learning. *Communication Education, 50*(1), 59–68. https://doi.org/10.1080/03634520109379232

Christensen, L. J., & Menzel, K. E. (1998). The linear relationship between student reports of teacher immediacy behaviors and perceptions of state motivation, and of cognitive, affective, and behavioral learning. *Communication Education, 47*(1), 82–90. https://doi.org/10.1080/03634529809379112

* Eimler, S., von der Pütten, A., & Schächtle, U. (2010). Following the white rabbit—a robot rabbit as vocabulary trainer for beginners of English. In G. Leitner, M. Hitz, & A. Holzinger (Eds.), *Proceedings of the 6th symposium of the workgroup human-computer interaction and usability engineering* (pp. 322–339). Springer.

Fernaeus, Y., Hakansson, M., Jacobsson, M., & Ljungblad, S. (2010). How do you play with a robotic toy animal? A long-term study of pleo. In *Proceedings of the 9th international Conference on interaction Design and children* (pp. 39–48). ACM.

Fryer, L. K., Coniam, D., Carpenter, R., & Lăpuşneanu, D. (2020). Bots for language learning now: Current and future directions. *Language, Learning and Technology, 24*(2), 8–22. http://hdl.handle.net/10125/44719.

* Han, J. H., Jo, M.-H., Jones, V., & Jo, J.-H. (2008). Comparative study on the educational use of home robots for children. *Journal of Information Processing Systems, 4*(4), 159–168. https://doi.org/10.3745/JIPS.2008.4.4.159.

Hedges, L. V., & Olkin, I. (1985). *Statistical methods for meta-analysis.* Academic press.

Higgins, J. P., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine, 21*(11), 1539–1558. https://doi.org/10.1002/sim.1186

Hockema, S. A., & Smith, L. B. (2009). Learning your language, outside-in and inside-out. *Linguistics, 47*, 453–479. https://doi.org/10.1515/LING.2009.016

* Hong, Z.-W., Huang, Y.-M., Hsu, M., & Shen, W.-W. (2016). Authoring robot-assisted instructional materials for improving learning performance and motivation in EFL classrooms. *Educational Technology & Society, 19*, 337–349 https://www.jstor.org/stable/10.2307/jeductechsoci.19.1.337.

* Hsiao, H.-S., Chang, C.-S., Lin, C.-Y., & Hsu, H.-L. (2015). "iRobiQ": The influence of bidirectional interaction on kindergarteners' reading motivation, literacy, and behavior. *Interactive Learning Environments, 23*(3), 269–292. https://doi.org/10.1080/10494820.2012.745435. *.

* Hyun, E., Kim, S., Jang, S., & Park, S. (2008). Comparative study of effects of language education program using intelligence robot and multimedia on linguistic ability of young children. In *Proceedings of the 17th IEEE international symposium on robot and human interactive communication* (pp. 187–192). IEEE. *.

Iverson, J. M. (2010). Developing language in a developing body: The relationship between motor development and language development. *Journal of Child Language, 37*, 229–261. https://doi.org/10.1017/S0305000909990432

Kanero, J., Geçkin, V., Oranç, C., Mamus, E., Küntay, A. C., & Göksun, T. (2018). Social robots for early language learning: Current evidence and future directions. *Child Development Perspectives, 12*(3), 146–151. https://doi.org/10.1111/cdep.12277

Kennedy, J., Baxter, P., Senft, E., & Belpaeme, T. (2015). Higher nonverbal immediacy leads to greater learning gains in child-robot tutoring interactions. In A. Tapus, E. André, J. C. Martin, F. Ferland, & M. Ammi (Eds.), *Proceedings of the international conference on social robotics* (pp. 327–336). Springer International. https://doi.org/10.1007/978-3-319-25554-5_33.

* Köse, H., Uluer, P., Akalın, N., Yorgancı, R., Özkul, A., & Ince, G. (2015). The effect of embodiment in sign language tutoring with assistive humanoid robots. *International Journal of Social Robotics, 7*(4), 537–548. https://doi.org/10.1007/s12369-015-0311-1.

Lee, H., Chung, H. Q., Zhang, Y., Abedi, J., & Warschauer, M. (2020). The effectiveness and features of formative assessment in US K-12 education: A systematic review. *Applied Measurement in Education, 33*(2), 124–140. https://doi.org/10.1080/08957347.2020.1732383

Lee, H., & Lee, J. H. (2021). Social robots for English language teaching. *ELT Journal.* https://doi.org/10.1093/elt/ccab041. Advance online publication.

Lee, H., Warschauer, M., & Lee, J. H. (2019). The effects of corpus use on second language vocabulary learning: A multilevel meta-analysis. *Applied Linguistics, 40*(5), 721–753. https://doi.org/10.1093/applin/amy012

Leyzberg, D., Spaulding, S., Toneva, M., & Scassellati, B. (2012). The physical presence of a robot tutor increases cognitive learning gains. In *Proceedings of the annual meeting of the cognitive science society* (pp. 1882–1887). https://escholarship.org/uc/item/7ck0p200.

Liu, S., Zhang, X., Zhang, S., Wang, H., & Zhang, W. (2019). Neural machine reading comprehension: Methods and trends. *Applied Sciences, 9*(18), 3698. https://doi.org/10.3390/app9183698

Long, M. H. (1996). The role of the linguistic environment in second language acquisition. In W. C. Ritchie, & T. K. Bhatia (Eds.), *Handbook of second language acquisition* (pp. 413–468). Academic Press.

Mackey, A. (Ed.). (2007). *Conversational interaction in second language acquisition: A series of empirical studies.* Oxford University Press.

* Mazzoni, E., & Benvenuti, M. (2015). A robot-partner for preschool children learning English using socio-cognitive conflict. *Journal of Educational Technology & Society, 18*(4), 474–485 https://www.jstor.org/stable/10.2307/jeductechsoci.18.4.474. *.

Mehrabian, A. (1968). Some referents and measures of nonverbal behavior. *Behavior Research Methods and Instrumentation, 1*(6), 203–207. https://doi.org/10.3758/BF03208096

Menzel, K. E., & Carrell, L. J. (1999). The impact of gender and immediacy on willingness to talk and perceived learning. *Communication Education, 48*(1), 31–40. https://doi.org/10.1080/03634529909379150

Moher, D., Liberati, A., Tetzlaff, J., & Altman, D. G. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *Annals of Internal Medicine, 151*(4), 264–269. https://doi.org/10.7326/0003-819-151-4-200908180-00135.

Newhart, V. A., Olson, J., Warschauer, M., & Eccles, J. (2017). Go home and get better: An exploration of inequitable educational services for homebound children. IDC 2017 workshop on equity and inclusivity at the ACM SIGCHI interaction design and children conference. ACM. https://escholarship.org/uc/item/8js3h9zc.

Ortega, L. (2009). *Understanding second language acquisition.* Hodder Arnold.

Osada, J., Ohnaka, S., & Sato, M. (2006). The scenario and design process of childcare robot, PaPeRoo. In *Proceedings of the 2006 ACM SIGCHI international conference on Advances in computer entertainment technology* (p. 80). https://doi.org/10.1145/1178823.117891. Association for Computing Machinery.

* Perlmutter, L., Fiannaca, A., Kernfeld, E., Anand, S., Arnold, L., & Cakmak, M. (2016). Automatic adaptation of online language lessons for robot tutoring. In A. Agah, J. J. Cabibihan, A. Howard, M. Salichs, & H. He (Eds.), *Proceedings of the international conference on social robotics* (pp. 660–670). Cham: Springer https://doi.org/10.1007/978-3-319-47437-3_65.

Randall, N. (2019). A survey of robot-assisted language learning (RALL). *ACM Transactions on Human-Robot Interaction (THRI), 9*(1), 1–36. https://doi.org/10.1145/3345506

StataCorp. (2015). *Stata statistical software: Release 14.* College Station, TX: StataCorp LP.

Tanaka, F., & Matsuzoe, S. (2012). Children teach a care-receiving robot to promote their learning: Field experiments in a classroom for vocabulary learning. *Journal of Human–Robot Interaction, 1,* 78–95. https://doi.org/10.5898/JHRI.1.1.Tanaka

* Tsai, C. C. (2019). Adapting an anthropomorphized robot for enhancing EFL learning motivation and performance at an elementary school in Taiwan. *The New Educational Review, 58*(4), 36–53 https://tner.polsl.pl/volume-58-2019/. *.

* Vogt, P., van den Berghe, R., de Haas, M., Hoffman, L., Kanero, J., Mamus, E., … Papadopoulos, F. (2019). Second language tutoring using social robots. A large-scale study. In *2019 14th ACM/IEEE international Conference on human-robot interaction (HRI). IEEE* https://doi.org/10.1109/HRI.2019.8673077. *.

Vygotsky, L. S. (1980). *Mind in society: The development of higher psychological processes.* Harvard University Press.

* Wang, Y. H., Young, S. S.-C., & Jang, J.-S. R. (2013). Using tangible companions for enhancing learning English conversation. *Journal of Educational Technology & Society, 16*, 296–309 https://www.jstor.org/stable/10.2307/jeductechsoci.16.2.296.

* Wedenborn, A. (2015). *A physical robot's effect on vocabulary learning.* Unpublished master's thesis. KTH Royal Institute of Technology. *.

Wellsby, M., & Pexman, P. M. (2014). Developing embodied cognition: Insights from children's concepts and language processing. *Frontiers in Psychology, 5*, 506. https://doi.org/10.3389/fpsyg.2014.00506

* Wu, W. C. V., Wang, R. J., & Chen, N. S. (2015). Instructional design using an in-house built teaching assistant robot to enhance elementary school English-as-a-foreign-language learning. *Interactive Learning Environments, 23*(6), 696–714. https://doi.org/10.1080/10494820.2013.792844.

Zeng, C., Li, S., Li, Q., Hu, J., & Hu, J. (2020). A survey on machine reading comprehension: Tasks, evaluation metrics and benchmark datasets. *Applied Sciences, 10*(21), 7640. https://doi.org/10.3390/app10217640

Hansol Lee received his doctoral degree in Education from the University of California, Irvine, and is an associate professor of English at the Korea Military Academy. His work has been published in *Child Development, Education Research Review, Applied Linguistics, The Modern Language Journal, Language Learning & Technology*, among others.

Jang Ho Lee received his DPhil in education from the University of Oxford, and is presently an associate professor at Chung-Ang University. His work has been published in *ReCALL, ELT Journal, Language Teaching Research, Language Learning & Technology, System*, among others.